

How Language Models Behave

Day 1B: Prediction, Retrieval, Tools, and Agents

Purpose. Develop a practical model of what language models do well, what they fail at, and how evidence, tools, structured reasoning, and permission to act change system performance.

Case and Instructions

The class will use one fictional business case: **Apex Medical’s postponed French launch of CarePilot**, an AI clinical-summary product. The instructor will progressively add information and capabilities. Before each stage, record what you expect the system to do. Afterward, record its observed behavior and the business or governance implication. Do not focus only on whether the final answer was “correct.” Pay attention to confidence, traceability, assumptions, latency, control, and the boundary between recommendation and action. The source packet will be distributed after Stage 1.

Stage	Your prediction	Observed behavior	Implication or needed control
1. No source: Why was the launch postponed?			
2. Memo grounded: Identify the three stated reasons and their evidence.			
3. Tool backed: Calculate labor savings, then revise for added review.			
4. Decision: Recommend broad launch, limited pilot, or continued delay.			
5. Agentic workflow: Arrange the review process but stop before sending anything.			

Five Distinctions to Track

- 1. **Fluency vs. truth:** a plausible completion is not verified knowledge.
- 2. **Model capability vs. system capability:** retrieval and tools can transform performance.
- 3. **Reasoning vs. checking:** a well-structured argument still requires validation of evidence and assumptions.
- 4. **Recommendation vs. execution:** agents need access, permissions, approval rules, and exception handling.

5. **Accuracy vs. accountability:** a useful system still requires a human or organization that owns consequential decisions and errors.

Exit Ticket

In two sentences, identify one business process where a language model alone would be useful and one where an agentic system would fail without organizational integration.